DES class activity

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The given first-round output is **00101011**. The second-round key is **10111001**. These are the simplified DES tables printed in the activity, which are different from the full DES tables used in HBEA.

Second round, step by step

1. Split and swap. L1 = 0010 and R1 = 1011. After the middle swap, L = 1011 and R = 0010.

2. Expand R. Using E/P = [4, 1, 2, 3, 2, 3, 4, 1], E/P(0010) = 00010100.

3. XOR with K2. 00010100 XOR 10111001 = 10101101. Split this into 1010 and 1101.

4. Apply S0. For 1010: outer bits 10 give row 2; inner bits 01 give column 1. S0[2][1] = 01.

5. Apply S1. For 1101: outer bits 11 give row 3; inner bits 10 give column 2. S1[3][2] = 01.

6. Apply P4. Join S0 and S1 to get 0101. P4 = [2, 4, 3, 1] gives 1100.

7. Update the left half. 1011 XOR 1100 = 0111. Keep R = 0010, giving 01110010 before the final permutation. There is no swap after the last round.

8. Apply inverse IP. Using [4, 1, 3, 5, 7, 2, 8, 6], the final result is 10101100.

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The ciphered text: 10101100
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Worksheet clarification. The activity text and its parameter box both give K2 = 10111001. The red diagram label has only seven bits (1011100), so the complete eight-bit value is used here. Rows and columns above are numbered from zero.

Verification. Run java -cp build DesWorksheet. It prints each intermediate value and checks the final answer. A separate direct calculation also produced 10101100.